# Predictive Modeling of Electronic Response in Low-Dimensional Interfaces

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## Research Vision

Interfaces increasingly determine the performance of electronic devices. This is especially true as device dimensions approach the atomic scale and incorporate low-dimensional materials. At these scales, interfaces become active quantum systems whose response to applied electric fields governs device operation[1,2]. The same physics is central to oxide ferroelectrics, where electric fields switch polarization, and 2D memristive materials, where field-driven charge, defects, or ions alter resistance states[3,4].

At the atomic scale, field response is the redistribution of electron density under an applied perturbation. Density-functional theory (DFT) provides this density and allows the response to be calculated from first principles. Capacitance connects this microscopic response to experiment because it measures how efficiently a system redistributes and stores charge. The capacitance–voltage ($C$–$V$) response therefore links electronic structure and field-induced charge redistribution to device behavior[5,6].

Predicting interfacial field response remains a major bottleneck in materials discovery. Classical electrostatic models neglect the electronic-structure limitations of low-dimensional materials, where quantum confinement and finite density of states add an energetic cost to charge accumulation[6]. Measurable field response therefore arises from coupled real-space charge redistribution and energy-space electronic structure, requiring a unified theoretical description.

My graduate research addressed this challenge by developing the Quantum Interfacial Capacitance (QIC) framework, a first-principles model that unifies electrostatic and quantum contributions to predict interfacial capacitance (Figure 1)[5]. Applied to graphene/$SiO_2$ interfaces, QIC reproduced experimentally observed nonlinear capacitance–voltage characteristics that conventional electrostatic or quantum-capacitance models could not explain, establishing the physical foundation for predictive modeling of low-dimensional interfaces.

Building on this foundation, my research vision is to develop scalable computational frameworks for predicting low-frequency field response in low-dimensional interfaces. Realizing this vision requires overcoming three bottlenecks:

- **Physical model generalization.** Extending physical models of interfacial field response across diverse low-dimensional materials and realistic interfaces.
- **Scalable field-response data generation.** Developing efficient workflows for large-scale interfacial datasets with field-response labels.
- **Transferable machine learning for electronic response.** Predicting electronic structure and charge redistribution beyond the interpolation regime required for materials discovery.

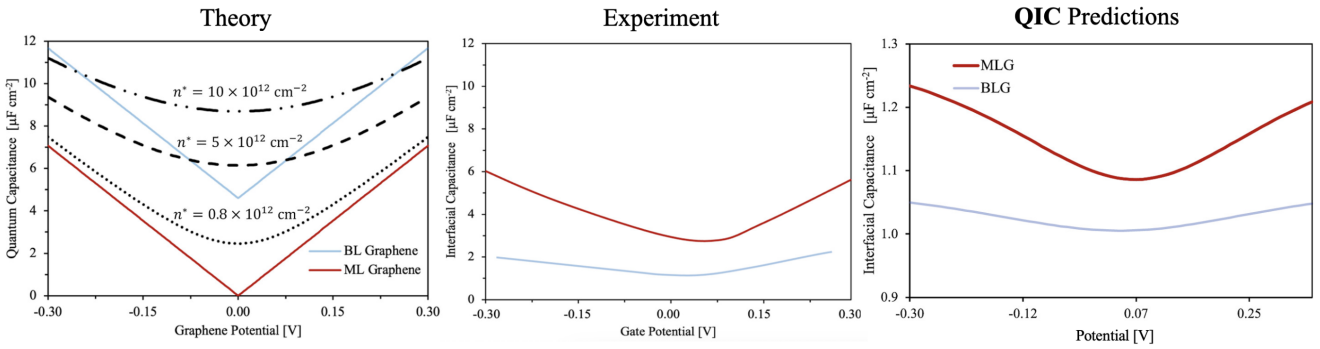



**Figure 1: The Quantum Interfacial Capacitance (QIC) framework bridges microscopic first-principles calculations and macroscopic device measurements.** Conventional quantum-capacitance theory fails to reproduce the experimental C–V response of monolayer (ML) and bilayer (BL) graphene, whereas QIC accurately captures the observed trends by combining quantum capacitance with first-principles electrostatic screening. This work establishes the physical foundation for the research program.

## Thrust I: Physics-Guided Modeling of Interfacial Field Response

The Quantum Interfacial Capacitance (QIC) framework established a unified electrostatic–quantum description of interfacial field response for graphene/$SiO_2$ interfaces. The goal of this thrust is twofold. **First, establishing the generality of the QIC framework** by validating its predictive capability across diverse classes of low-dimensional interfaces against available experimental measurements. **Second, identifying the minimum physical description required** to accurately predict interfacial field response by determining which physical ingredients can be simplified without sacrificing predictive accuracy.

These objectives are intrinsically linked. Systematic validation establishes the physical ingredients governing interfacial field response, while physics-guided model reduction identifies the minimum computational fidelity required to preserve them. Together, they provide the physical foundation for scalable high-throughput simulation by maximizing computational efficiency without compromising predictive accuracy.

Figure 2 illustrates the type of first-principles electrostatic information underlying this thrust: an atomistic graphene/silica interface under an applied field, together with the planar charge-density and electrostatic-potential response that connect microscopic electronic structure to macroscopic field response.

**Model validation and generalization.** The QIC framework will be evaluated across diverse low-dimensional interfaces, including graphene, transition-metal dichalcogenides (TMDs), MXenes, and other experimentally characterized systems. These studies will establish the generality of the framework, identify its limitations, and reveal which physical mechanisms must be incorporated to accurately describe different classes of interfaces.

**Physics-guided model reduction.** Systematic benchmark studies will determine the minimum computational fidelity required to preserve experimentally relevant trends in interfacial field response. Particular emphasis will be placed on understanding the sensitivity of field-induced charge redistribution and electronic density of states to first-principles fidelity, identifying when lower-cost approximations, simplified electronic-structure models, or reduced physical descriptions

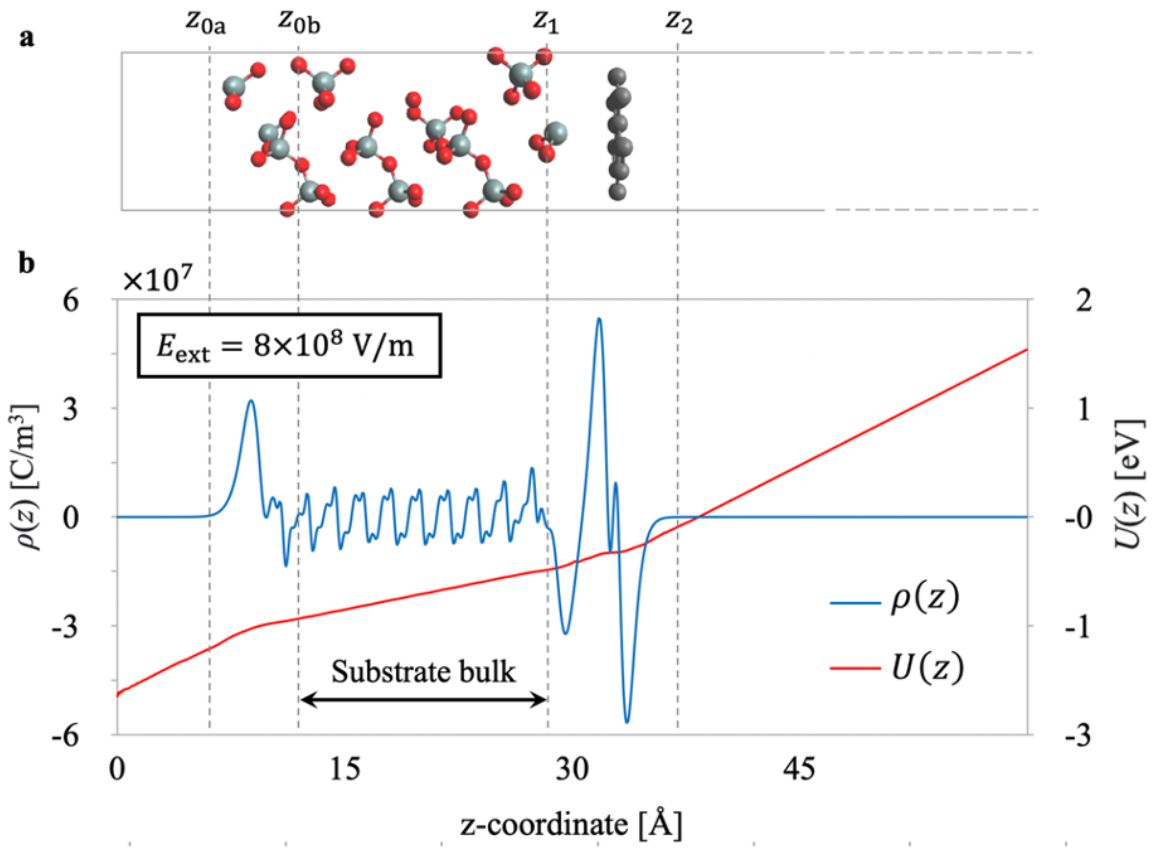



**Figure 2: First-principles field response of a graphene/silica interface.** Atomistic interface geometry and planar profiles of charge density $\rho(z)$ and electrostatic potential $U(z)$ under an applied external field. This representation connects the microscopic structure of the interface to the charge redistribution and electrostatic screening quantities used in physics-guided model reduction.

remain valid. The resulting guidelines will define the optimal balance between physical accuracy and computational efficiency, providing the foundation for the scalable simulation workflows developed in the next research thrust. The resulting first-principles quantities can also parameterize atomistic or MLIP models and, through collaboration, phase-field models that connect electronic structure to polarization domains and switching.

## Thrust II: Scalable Generation of Interfacial Field-Response Datasets

The physical models developed in Thrust I can only enable materials discovery if they are translated into scalable datasets of interfacial field response. While recent atomistic machine-learning models have transformed structure relaxation and energy prediction, comparable datasets of electronic response remain largely unavailable. Systematic first-principles datasets of field-induced charge redistribution, electronic density of states, and derived interfacial observables across diverse low-dimensional interfaces are still missing.

This thrust will develop physics-aware computational workflows that integrate existing atomistic databases, foundation models, and first-principles simulations to generate these missing datasets at scale. Rather than applying a fixed computational protocol to every interface, the workflow will adaptively select the lowest level of theory consistent with the physical fidelity established in Thrust I, enabling efficient generation of standardized interfacial field-response datasets.

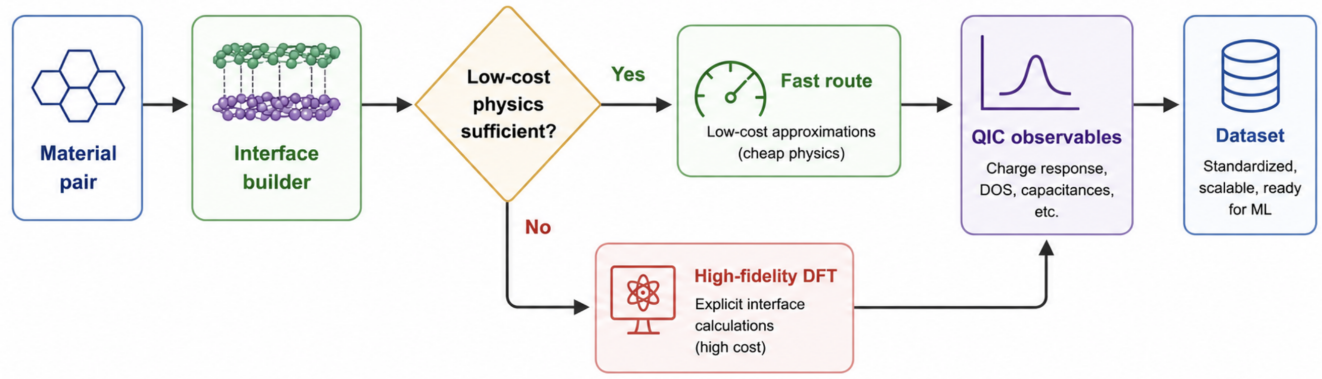



**Figure 3: Physics-aware adaptive workflow for interfacial response datasets.** Material pairs are converted into interfaces, routed through low-cost or high-fidelity calculations, and standardized into QIC-relevant observables for machine learning.

Different interfaces require different levels of computational fidelity. Weakly interacting systems may only require isolated-layer electronic-structure calculations, whereas strongly hybridized interfaces demand explicit first-principles treatment. Similar adaptive decision strategies will be developed for structure relaxation, electronic-structure calculations, and charge-response simulations, while quantifying the uncertainty introduced by each approximation (Figure 3).

The resulting framework will transform interfacial field-response calculations from manually curated simulations into scalable, standardized data-generation pipelines, providing the computational infrastructure required for machine-learning-driven materials discovery.

## Thrust III: Physics-Guided Machine Learning for Interfacial Electronic Response

This thrust develops machine-learning models for the two quantities most central to interfacial field response: charge redistribution under an applied field and the electronic density of states. The goal is to determine how physical representations, target design, and learning strategy control transferability.

I view scientific machine learning as a form of physical modeling. A useful model should expose the features that control the target without introducing unnecessary degrees of freedom that promote overfitting. Similarly, the target itself should be chosen so that the mapping from structure to response is as smooth and transferable as possible within the model's inductive bias. This perspective motivates three linked directions.

**Representation design.** Electronic-response models will compare geometric, electronic-structure-informed, and hybrid representations. For local or near-local response quantities, atomistic graph representations may be sufficient. For global quantities such as the density of states, richer inputs such as Hamiltonian-informed descriptors may expose electronic couplings that are poorly represented by geometry alone (Figure 4)[7–9].

**Target design.** Electrostatic response and electronic structure will be treated as separate learning problems. For electrostatic response, targets may include field-induced charge redistribution, laterally averaged response profiles such as $\Delta\rho(z)$, electrostatic capacitance, or reduced descriptors identified in Thrust I. Existing charge-density models provide an important starting point, but this

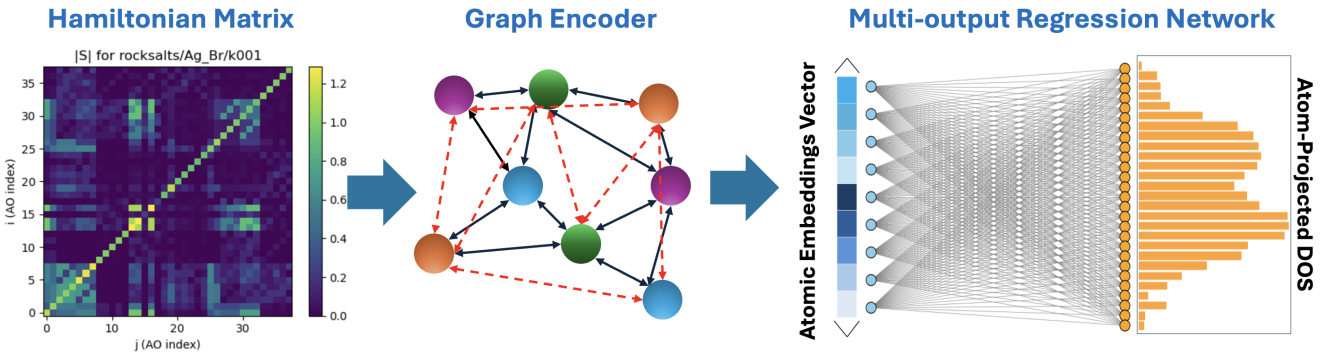



**Figure 4: Hamiltonian-informed learning of electronic structure.** The Hamiltonian matrix encodes nonlocal electronic couplings that are not naturally captured by purely local atomistic message passing. This thrust will investigate how Hamiltonian-inspired and graph-based representations can be used to predict global spectral quantities such as the density of states, $D(E)$, for transferable interfacial electronic-response modeling.

thrust will test whether response-specific targets generalize more reliably for interfaces[10–15]. For electronic structure, the primary target is the absolute density of states, $D(E)$. This separation is important because charge redistribution may be highly field-sensitive, while $D(E)$ may be comparatively invariant under moderate applied fields.

**Physics-informed learning strategies.** Delta learning and hierarchical learning will be used to improve transferability when direct learning of the full target is inefficient. In delta learning, a low-cost physical model captures part of the target, while ML learns the residual needed to approach first-principles accuracy. In hierarchical learning, a geometry-based model can provide an initial prediction, while a second model using richer physical representations corrects the remaining error. These strategies are designed to make the learned target smoother and better aligned with the information available to the model.

Together, these studies will establish design principles for learning interfacial electronic response with limited data, emphasizing physically meaningful features and targets rather than brute-force scaling alone.

## Integration of the Research Program

The three research thrusts are designed to progress in parallel while continuously informing one another. The physics thrust establishes transferable physical models and computational approximations for predictive interfacial field-response calculations. These insights guide the high-throughput simulation thrust, which transforms the underlying physics into scalable computational workflows and standardized datasets of interfacial electronic response. These datasets, in turn, enable the machine-learning thrust, which develops predictive surrogate models for electronic response while identifying improved physical representations and computational simplifications that feed back into both the physics models and simulation workflows.

Together, these thrusts establish a closed-loop research program that integrates physical understanding, scalable computation, and machine learning to enable predictive virtual screening of low-dimensional interfaces before experimental fabrication.

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